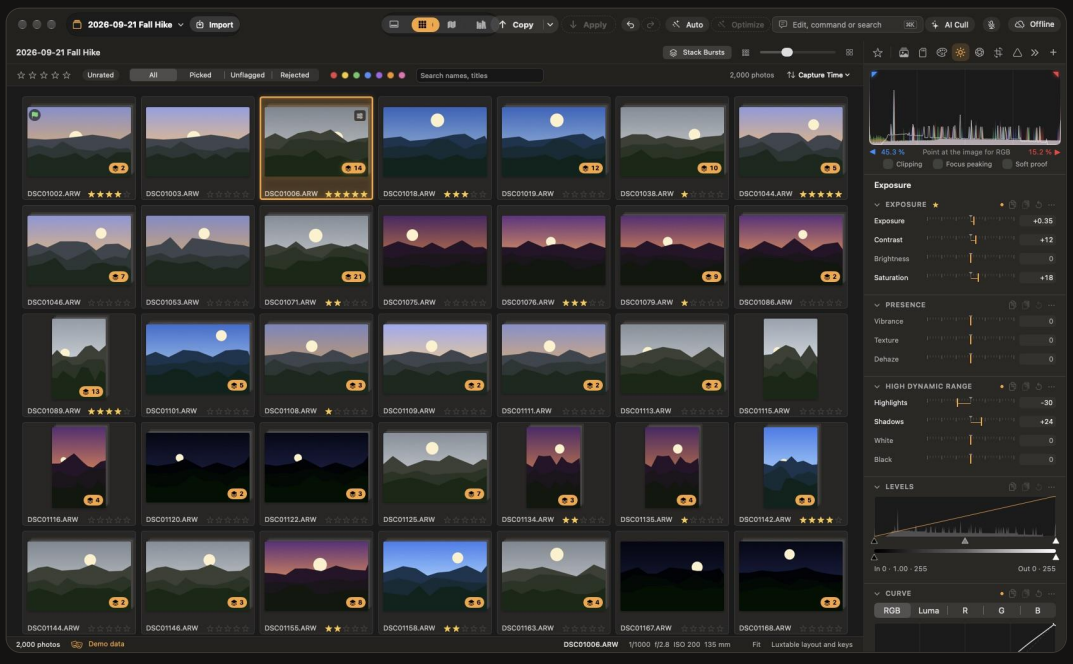
User Guide

Import, cull, develop, export and archive your RAW photos on a Mac



Luxtable is free software under the Apache License 2.0 · The Luxtable Authors

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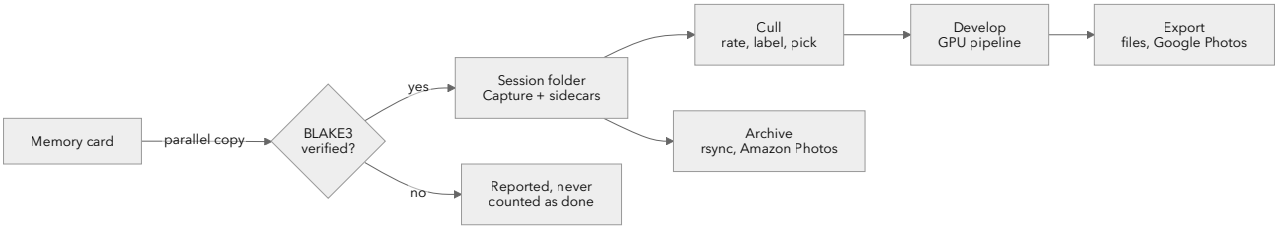
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CHAPTER 1

Welcome to Luxtable

Luxtable is a RAW photo workflow app for macOS. It takes you from a memory card to finished, backed-up photos: import and verify, cull, develop, export and archive. It is built around three ideas.

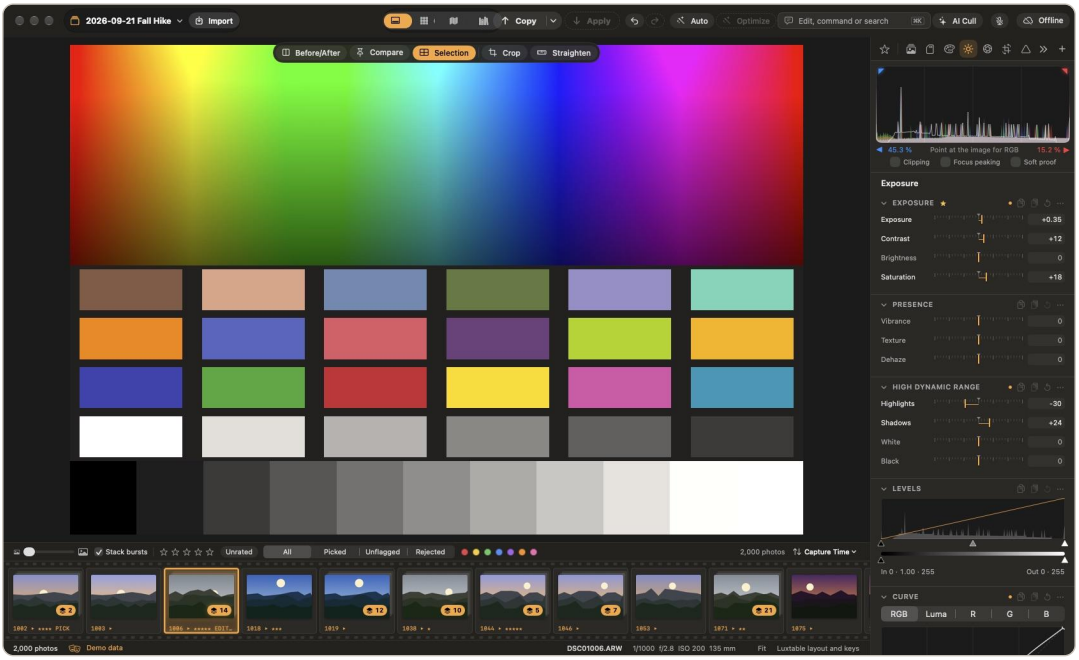
- **Speed where you spend your time.** Thumbnails appear straight away from the preview each RAW file carries, and full renders fill in behind them. Culling is driven by single keys, and the develop pipeline runs on the GPU, so sliders respond as you drag.
- **Sessions you own.** A session is an ordinary folder: your originals, plus a small sidecar file per photo that holds its rating, labels, keywords and edits. The sidecars are the source of truth. Luxtable keeps a database alongside them only to be fast, and can rebuild it from the sidecars at any time. Library mode still shows every session's photos together.
- **Never losing a file.** Imports are checked byte for byte before they count as done. Luxtable never deletes anything from a memory card or from your archive, and every file it writes is written in full before it replaces the old one.



From the card to the keepers.

The main window

The main window has a toolbar at the top, the browser in the middle and the tools beside it.



The main window in demo mode: the viewer, the filmstrip below it and the Exposure tools.

- **Toolbar:** the session menu, Import, the view switch (**Viewer**, **Grid**, **Map**, **Library**), the **Edit with words** field (`⌘K`), archive status and the **Copy** / **Apply** buttons.
- **Browser:** the viewer with a filmstrip, or a grid of thumbnails. The filmstrip can sit at the bottom or as a column on the left or right (**View** ▸ **Filmstrip**).
- **Tools:** tabs of tool panels (Library, Import & Archive, Color, Exposure, Lens, Composition, Details, Layers, Metadata, Output, and your Favorites). **View** ▸ **Tools on Left** moves them to the other side.

Press `?` at any time for a cheat sheet of the keys, and `F` or `Tab` to hide the panels and see only the photo.

CHAPTER 2

Install and first run

Requirements

- A Mac with Apple silicon, running macOS 26 or later.
- To build it: the Xcode Command Line Tools (Swift 6.2) and Rust 1.90 or newer. Xcode itself isn't needed.

Download

Download `Luxtable-X.Y.Z-macos-arm64.dmg` from the project's GitHub releases, open it and drag Luxtable.app onto the Applications folder in its window. The app is self-signed (Luxtable's own certificate), not notarised: the first time, right-click Luxtable.app and choose **Open**. From then on Luxtable keeps itself up to date (see Updates).

Build from source

You can also build Luxtable from its source code. In Terminal:

```
git clone https://github.com/sbaillar/luxtable.git
cd luxtable
LT_SKIP_MODELS=1 scripts/build-app.sh --run
```

This builds `dist/Luxtable.app` and opens it. You can then copy it to your Applications folder.

- `LT_SKIP_MODELS=1` builds faster by leaving out the AI Denoise model (building it needs Python 3.13 and a download). Without it, AI Denoise shows as unavailable. Leave the variable out for a full build.
- `scripts/make-signing-identity.sh` creates a local signing identity once, so macOS remembers the permissions you give Luxtable across rebuilds.

Try it without your photos

Demo mode opens sample sessions of synthetic photos and simulates imports and archiving. It never reads or changes your own files:

```
dist/Luxtable.app/Contents/MacOS/Luxtable --demo
```

Every screenshot in this guide was taken in demo mode.

First run

The first time Luxtable opens, a short setup sheet walks you through a few steps. **Skip for Now** keeps the defaults, and everything can be changed later in Settings (`⌘,`).

1. **Welcome:** what Luxtable does.

2. **What did you use before?** Choose Luxtable, Lightroom Classic or Capture One. This only moves panels, keyboard shortcuts and a few names to where your hands expect them (Settings ▸ Familiar). If you are new to photo editing, choose **Nothing yet: I'm new to photo editing** to start in Simple mode: an Auto button, looks and a few easy sliders.
3. **Where should sessions live?** The folder where new sessions are created. Pick your fastest disk.
4. **Archive (optional):** tick **Archive sessions automatically** and choose a folder (for example on a NAS) that sessions are copied to in the background. It stays off until you turn it on.
5. **Amazon Photos (optional):** a folder the Amazon Photos app backs up. This step only appears while the Amazon Photos integration is on (see Optional integrations).

Permissions

macOS asks before Luxtable can read removable volumes, network volumes or folders such as Pictures. Allow these for the drives you keep photos on. Voice commands ask for microphone and speech recognition access the first time you turn them on.

The RAW helper

Luxtable's own decoders read DNG (including iPhone ProRAW), Sony ARW, Canon CR2, Nikon NEF and Pentax PEF. Most other RAW formats, and the camera data that ARW, NEF and CR2 files need to develop, come from the optional **RAW helper**, a separate program you install only if you need it. Without it, those photos import and cull from their embedded previews, and HDR and panorama merges aren't available.

Open **Settings ▸ RAW ▸ Download RAW Helper**, or click **Get RAW Helper** on a photo that needs it. Luxtable checks the download against the fingerprint built into the app before it installs it. **Remove RAW Helper**, in the same place, takes it out again.

Updates

Luxtable looks for a new version shortly after it starts and then about once a day. When there is one, a small notice says so; it never interrupts you with a dialog in the middle of culling. **Luxtable ▸ Check for Updates…** checks straight away.

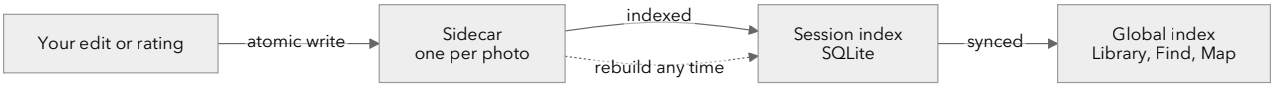
- The update sheet shows what's new. **Download & Restart** downloads the update, checks it, quits the normal way (pending changes are saved first) and reopens on the session and photo you were on, with the same selection, view and tool tab.
- **Download** installs it the next time you quit, **Later** reminds you another time, and **Skip This Version** stops the reminders for that version.
- The version it replaces is kept in Luxtable's Application Support folder, in **Previous Version** (only the last one), so you can go back by dragging it to Applications.

Settings ▸ Updates turns **Check for updates automatically** off, and has **Include beta versions**: betas are early builds of the next version, for trying new features before they're released, and may have bugs. Once that version is released, you are offered the release. The check sends only the app's version to GitHub.

CHAPTER 3

Sessions

Everything in Luxtable happens in a session: a folder of photos and their sidecars. Each import usually becomes a new session, but you can import into an existing one, or open any folder of photos as it is.



Your edits are written to the photo's sidecar. The indexes that make browsing and Library fast are built from the sidecars and can be rebuilt.

New, open and close

- **New Session** (**File ▸ New Session…**, ⇧⌘N): give it a name and a location. It is created with Capture, Selects, Output and Trash folders, ready for an import or for photos dragged in.
- **Open** (⌘O): choose a recent session, or **Choose Folder…** to open any folder. **File ▸ Open Recent** lists recent sessions; one on a drive that isn't connected is shown greyed out with the reason. **Browse All Sessions in Library** in the same sheet opens the Library instead.
- **Close Session** (⌥⌘W): closes it. Nothing is deleted, and it stays in the Sessions list.

Folders opened in place

When you open a folder that isn't a Luxtable session, Luxtable works on it in place: it never moves, renames or changes your files, and writes only its own sidecars and index. Renamed or edited files are recognised again when you reopen the folder. Commands that would move files, such as moving rejects to the session Trash, are turned off for such folders and say why.

What is in a session folder

```
My Session/
  Capture/      originals
  Selects/      picks moved out of Capture
  Output/       exports, unless a recipe says otherwise
  Trash/        the session trash (restorable)
  .lt/          sidecars (the source of truth) and the rebuildable index
```

Because a session is self-contained, you can move it to another drive, copy it to another Mac, or archive it, and it opens the same way.

A folder opened in place has just one `.lt/` folder at its top, with the same contents; your photos stay where they are.

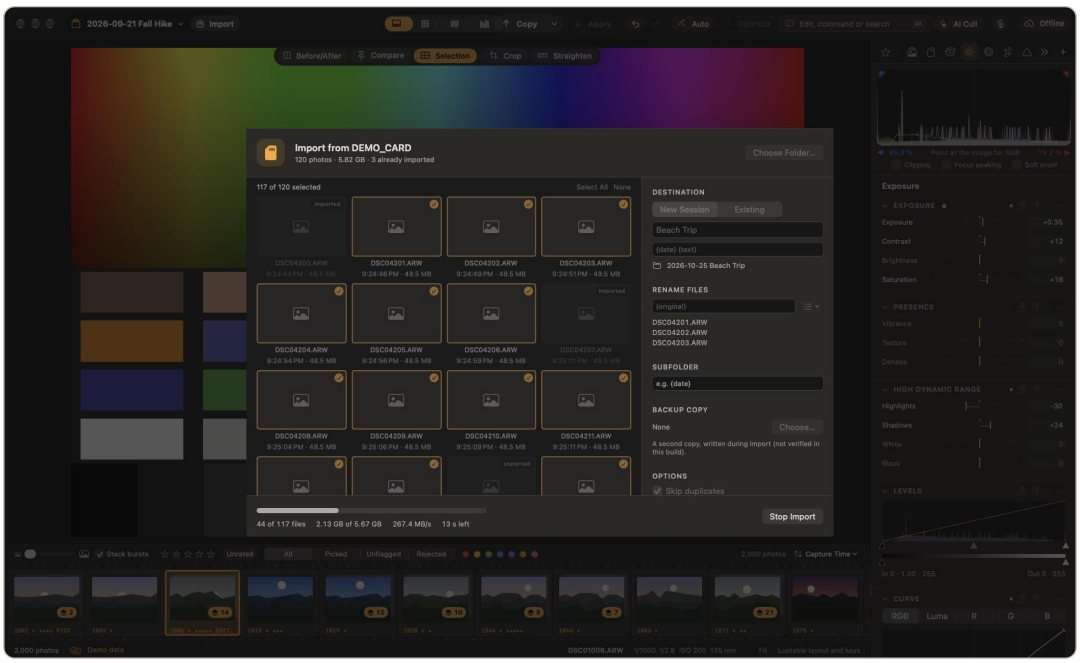
Disconnected drives and missing files

- Sessions on a drive that isn't connected stay listed, marked as disconnected. Nothing is hidden or removed.
- If an original has been moved or renamed, Luxtable finds it again by its content. **Locate…** and **Locate Folder…** point it to files you moved yourself.
- **Remove Missing** is only offered when Luxtable can prove the file is really gone from the right disk.

CHAPTER 4

Importing from a memory card

Insert a card and choose **File ▸ Import from Card…** (⇧⌘I) or the **Import** button. The import sheet lists the photos on the card. You can also import from any folder (**Choose Folder…**), or **Browse Card** to look at a card read-only without copying anything.



An import in progress (demo mode simulates the card).

Options

- **Session:** a new session (named for you, or by you) or an existing one.
- **Subfolder** inside the session's Capture folder, with tokens such as `{date}`, `{time}`, `{camera}` and `{text}`.
- **Rename Files** with a template: `{original}`, `{date}`, `{time}`, `{seq}`, `{seq4}`, `{camera}`, `{text}`.
- **Skip duplicates:** files already in any session are not copied again.
- **Backup Copy:** a second destination that receives the same verified files.
- **Re-read the card after copying:** reads every file from the card a second time and compares it. This catches a card reader that returned bad data once, and takes twice as long.

What makes it safe

- Files are copied in parallel, and each copy's BLAKE3 checksum is verified before the import counts it as done.
- Nothing is ever deleted from the card. When the import finishes you can eject the card safely from the sheet.

- If an import is interrupted (the card is pulled, the Mac sleeps), Luxtable offers to resume it the next time, and unfinished copies are never mistaken for complete ones.
- Metadata presets (for example your name and copyright) can be applied to every import, and the camera's own artist and copyright fields are read.

Photos from Capture One, Lightroom Classic, Aperture and iPhoto come in another way: see Coming from other apps.

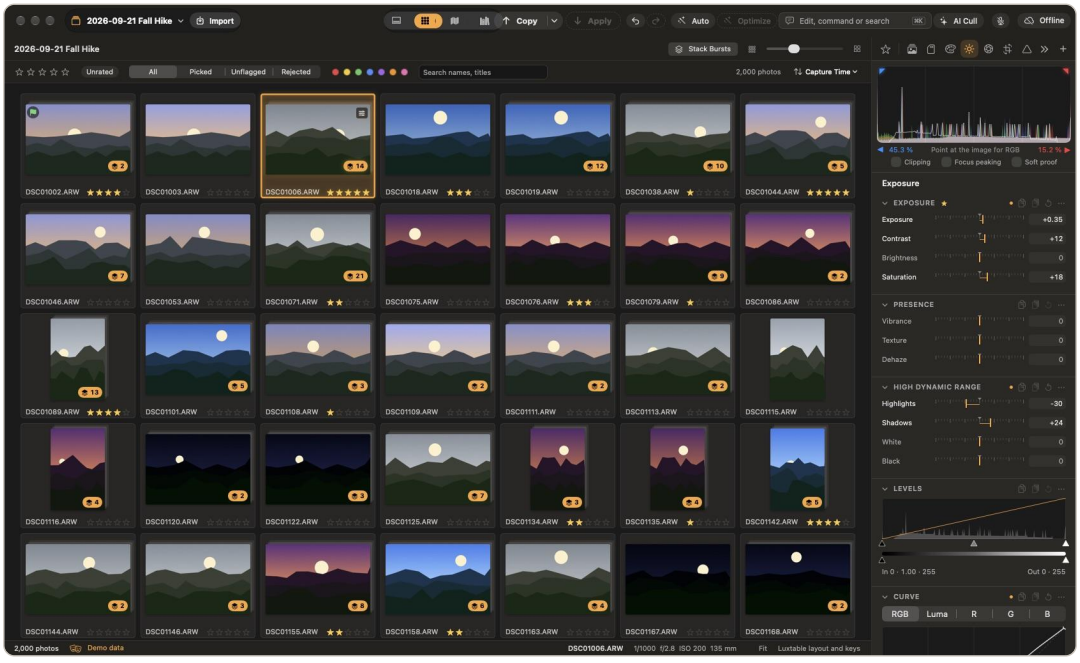
CHAPTER 5

Culling

Culling is choosing the keepers. Luxtable is built so you can do it with one hand on the keyboard: move with the arrows, and rate, label or flag with single keys. Each key can be changed in Settings ▸ Shortcuts.

Views and moving around

- G shows the grid, E (or ↩) the viewer. ⌃⌘F goes full screen.
- ← → move to the previous or next photo; ↑ ↓ move a row in the grid or in a filmstrip with several rows. Hold ⇧ to extend the selection.
- Space zooms to 100% at the cursor and back. ⌥⌘0 fits, ⌥⌘1 shows actual size, ⌘+ / ⌘– zoom.
- B hides the filmstrip. Drag the line beside it to show more rows or columns of thumbnails, or right-click a thumbnail to move the filmstrip to the left or right.
- The size slider in the filmstrip's filter bar, or **View ▸ Larger / Smaller Filmstrip Thumbnails** (⌃⌘= / ⌃⌘–), makes the filmstrip's thumbnails bigger or smaller. Larger thumbnails load sharper previews.
- View menu aids: Clipping Warning (⌥⌘J), Focus Peaking (⌥⌘P), Loupe (⌥⌘L) and Navigator (⌥⌘N). The same Clipping and Focus peaking switches are in the histogram panel (see The histogram panel).



The grid. Ratings, colour labels and pick or reject flags show on each thumbnail.

Ratings, labels and flags

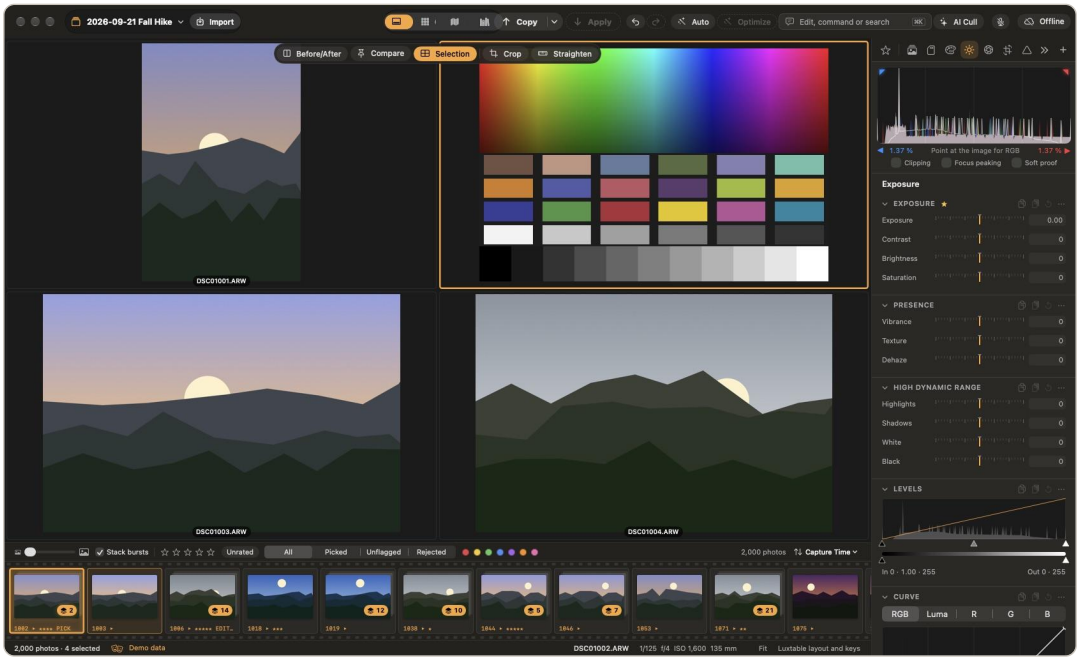
KEY	DOES
1 – 5, 0	Rate 1 to 5 stars; clear the rating

KEY	DOES
6 7 8 9 –	Red, yellow, green, blue or purple label (orange and pink exist but have no key until you assign one)
P / X / U	Pick, reject, unflag

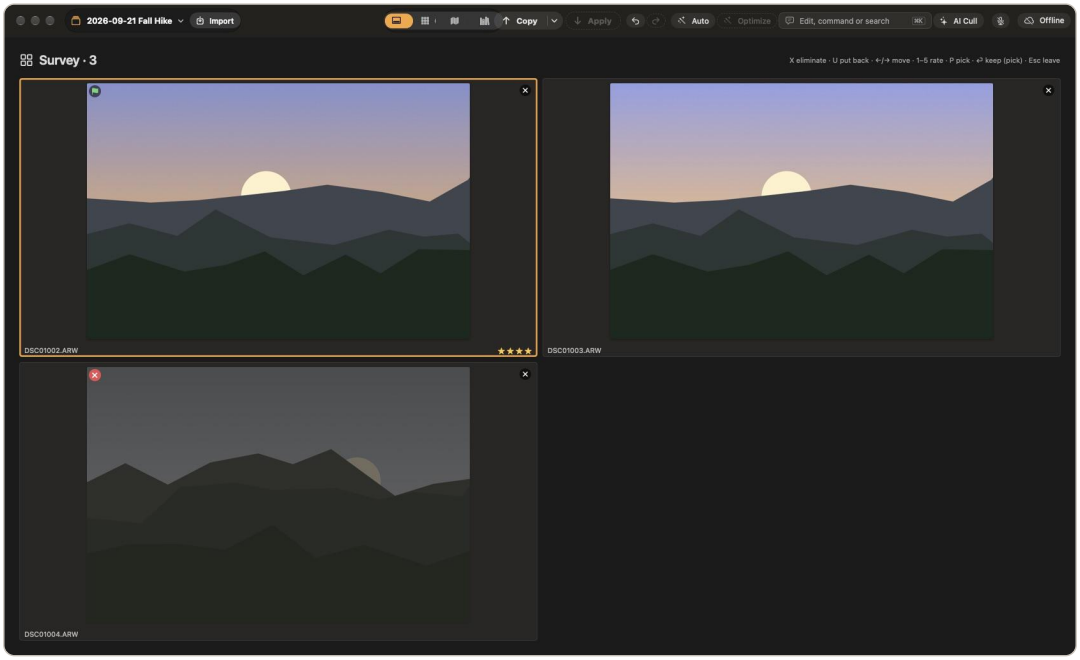
Photo ▸ Auto-Advance After Rating moves to the next photo after each rating. Ratings and labels apply to every selected photo.

Comparing

- **Multi-view:** select several photos (⌘-click or ⇧-click in the filmstrip) and the viewer shows them all, as large as they fit: two side by side, four as 2×2, up to twelve. The focused photo is outlined, and it is the one you zoom and edit. Turn this off with **View ▸ Show Selection in Viewer**.
- **Survey** (N or ⌥⌘S): the selected photos side by side. Press X to take the weakest out until only the keepers are left.
- **Before / after** (Y) splits the viewer between the original and your edit. **Pin for compare** (K) keeps a photo on screen while you move through the others.



Multi-view: four selected photos, the focused one outlined.



Survey: one photo already taken out with X.

Burst stacks

A burst is a run of frames shot in quick succession. Luxtable can show each burst as one stack, so you cull the moment once instead of frame by frame.

- **Stack bursts:** tick the **Stack bursts** checkbox beside the filmstrip's thumbnail size, or choose **View ▸ Stack Bursts**. It is on by default. Each burst then shows as one stack in the filmstrip and the grid, and the arrow keys step over it. Untick it to see and walk every frame.
- S expands or collapses the focused stack. With stacking off, S turns it on, with every burst collapsed.
- **Which frame is the cover:** the frame with the highest star rating wins, so rating a frame above the cover makes it the cover at once. Rejected frames are never the cover (unless every frame is rejected). When several frames share the top rating, the one you chose wins; if you haven't chosen, the AI's pick from the last AI Cull (then its quality score) decides; otherwise the earliest frame. With nothing rated, every frame ties.
- **The burst strip:** click a stack (or its badge) in the filmstrip and its frames open in a strip just above the filmstrip, highest rated first, each with its stars. Click a frame, or use ← →, to see it in the viewer; the rating and flag keys still apply to it.
- The cover is marked in the strip: **Your pick** or **AI pick** when it broke a tie, so you can see whose choice it was and change it.
- **Use as Cover** (the button, a frame's context menu, or ↩) is offered on the frames tied for the top rating. On the others the strip says what it takes, for example "Rate it ★★★ or higher to make it the cover".
- Esc, a click outside the strip or a click on the badge again closes the strip.
- **Your choice is kept:** it is saved with the session, so it is the same on every Mac that opens it. If a frame is later rated higher, that frame becomes the cover, but your choice is remembered and comes back when the ratings tie again. If the burst changes (an import or a delete), the choice stays while its frame is still in the burst. Survey starts on your chosen cover too.

AI Cull

Adjustments ▸ AI Cull (⇧⌘K) scores every photo for sharpness, exposure and closed eyes, and groups near-duplicates, all on this Mac. Use the scores to filter, then make the final call yourself.

When AI Cull rejects a photo, it says why in a short note, such as "Eyes closed" or "Duplicate of DSC00001.ARW: that frame is sharper". The note shows as **AI reject** in the info panel, in the viewer and in the reject badge's tooltip. Changing the flag yourself clears it.

Voice culling

Turn on **View ▸ Voice Commands** (⌃⌘V) and say what to do: "pick", "reject", "three stars", "next", "next 3", "go to photo 12", "reject the next 3", "rate the next 5 three stars", "show picks" or "undo 3 times". Say "what can I say?" for every phrase. Speech is recognised on this Mac. Settings ▸ Voice adds your own phrases (for example "keeper" for Pick).

Finishing a cull

- **Photo ▸ Move Rejected to Trash** moves rejects into the session's Trash folder. **Restore from Session Trash** brings them back.
- **Photo ▸ Move Picked and Rated to Selects…** moves the keepers into Selects.
- **Photo ▸ Delete Rejected…** asks before it deletes anything, and never touches a card or the archive.

Smart folders, filters and metadata

- **Smart Folders** (Library tab) collect photos by exact rules on rating, label, flag and keyword. ⌘0 shows all photos and ⌘1 – ⌘5 show only that many stars.
- **Filters** narrow the browser to picks, rejects, unflagged photos and more.
- **Metadata** (Metadata tab): title, caption, headline, creator, credit, copyright, location, instructions and usage terms (IPTC Core), and **Keywords**, for one photo or the whole selection. Fields that differ across the selection show as "Mixed" and are kept unless you change them.

CHAPTER 6

Library, Find, People and Map

Library mode

Library mode is Luxtable's catalog: the photos of every session in one fast grid, without opening each session. Open it in any of these ways:

- Choose **Library** in the toolbar's view switch.
- **File ▸ Browse All Sessions in Library** (⇧⌘L), or **Browse All Sessions in Library** in the Open Session sheet.
- To start Luxtable in the Library, set **Settings ▸ General ▸ Open in** to **Library (catalog)** (the default is **Session**). No session is opened at launch then; the one that would have opened opens when you leave the Library.

In the Library:

- Browse **All Photographs**, by session, by date (year, month, day) or by collection.
- **Opened Sessions** shows the photos of every session you have opened, as one. ⌘-click sessions in the sidebar to see just those together.
- Filter by rating, label, flag, camera and keyword; search file names, titles and keywords; sort by capture time, rating, file name or session.
- Rate, label and flag from the Library. The change is written to the photo's own session.
- **Collections** gather photos from any sessions. **Smart collections** keep a filter (**Smart Collection from Filter…**).
- Double-click a photo, or choose **Open in Session**, to open it in its session for editing.
- Sessions on a disconnected drive stay listed; their photos say the original isn't reachable.
- Photos load as you scroll, and the details of pages far from view are let go, so even a very large Library opens quickly and stays light.

Find

Edit ▸ Find… (⌘F) searches across every session in plain words, such as "pics of alex at the beach last summer". It understands the people you have named, places, dates, ratings and descriptions of what is in the photo, and returns both sessions and folders and individual photos, each with the reason it matched. Photo search runs on this Mac with a downloaded on-device model.

People

The **People** panel (Library tab) groups faces found in the background on this Mac. Answer "Who is this?" to name a group, merge groups that are the same person, or split off faces that are someone else. Named people can then be found with Find and used for people masks when editing. Nothing about faces leaves your Mac.

Map

Choose **Map** in the view switch to see where photos were taken, from the camera's GPS or an XMP sidecar, as clustered markers for this session or all sessions.

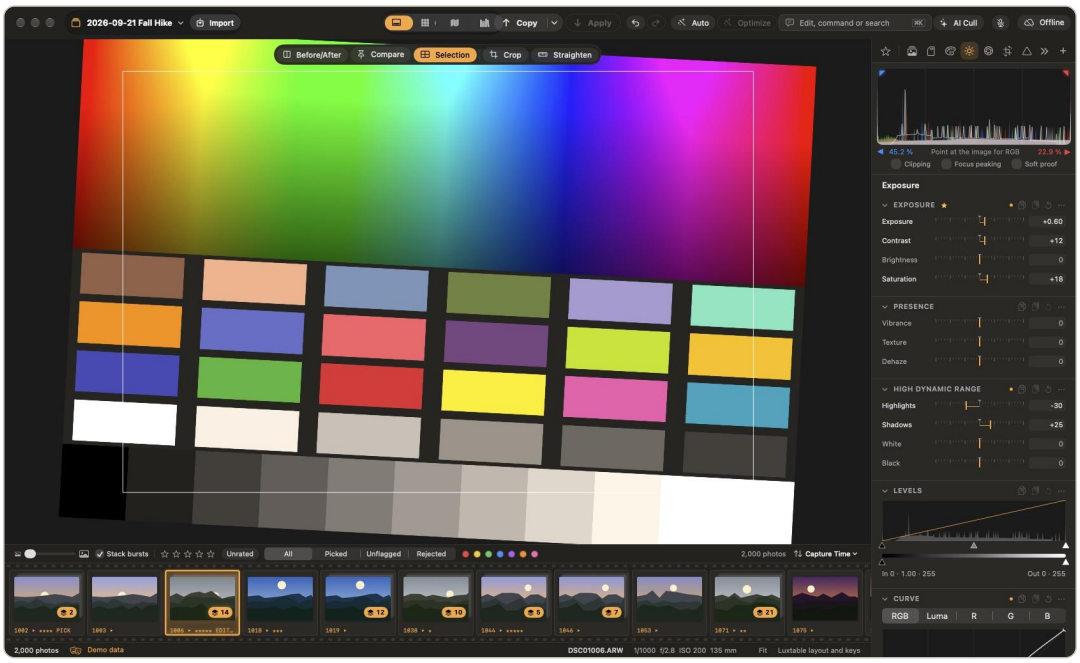
- Click a marker, **Draw Area**, or use **Filter to View** to show only the photos in that part of the map.
- Drag photos from the filmstrip onto the map to set their location. It is kept in the sidecar, never written to the RAW file, and **Remove Location** takes it off again.
- Map tiles come from Apple Maps, so the areas you look at are requested from Apple. Luxtable asks once before showing the map, and Settings ▸ General turns it off. Your photos and their locations are never uploaded.

CHAPTER 7

Developing

Edits are never applied to the original file. They are stored as settings in the sidecar and rendered by the develop pipeline on the GPU, so you can change or undo them at any time (⌘Z / ⇧⌘Z).

Adjustments ▸ Reset All Adjustments (⇧⌘R) takes the photo back as shot, with no adjustments, in one undo step. It does the same as the **Original** row in Styles & Presets.



A live edit: warmer white balance, more exposure, a crop and a straighten, with the histogram.

Simple mode

New to photo editing? **Simple mode** swaps the tool tabs for one panel with just the essentials. Switch with **Simple | Full** at the top of the tools pane, **View ▸ Simple Mode**, or ^⌘S. Luxtable remembers your choice.

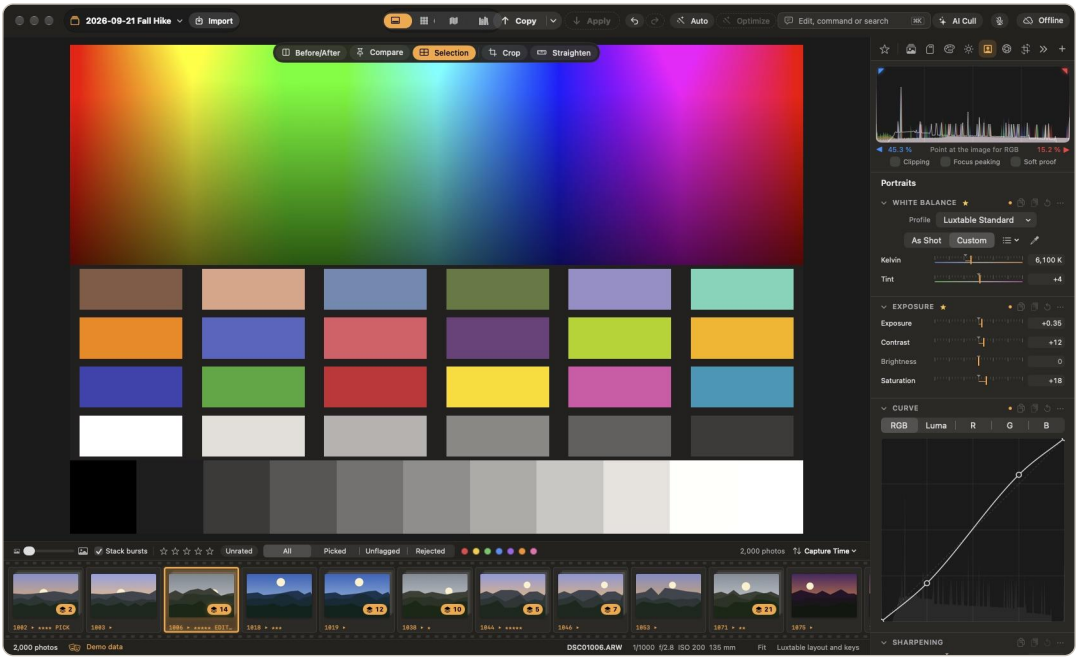
- **Auto** fixes brightness, contrast and colour in one click (the same as Auto Adjust, ⇧A).
- **Looks** shows your photo in each look. Hover a look to see it on the photo; click to use it. **Original** goes back to the photo as shot. An **Intensity** slider appears under the strip so you can make a look stronger or softer.
- Six sliders: **Light** (brighter or darker), **Contrast**, **Color** (stronger or softer colours; all the way left is black and white), **Warmth** (cooler or warmer light; 0 is the camera's own), **Highlights** (bring back a bright sky) and **Shadows** (show detail in dark areas). Double-click a slider to put it back to 0.
- **Crop**, **Straighten** and **Rotate Left**, and **Before / After** to compare with the original (or press Y).
- **Start Over** takes the photo back to how it was shot. ⌘Z undoes any step.

Simple mode is not a separate kind of edit. Each slider is one of the regular tools under a plainer name (Light is Exposure, Color is Saturation, Warmth is the white balance temperature), so when you switch to **Full** your changes are there, and anything you change in Full mode shows in Simple mode too. Choosing a tool from the **Tools** menu or **View ▸ Show History** switches to Full.

Tool tabs

Tools live in tabs across the top of the tools pane. Tabs that don't fit go in a » menu.

- The + button adds or removes any tool in one click. Each panel's ⋯ menu removes or moves it, and panels can be dragged into any order.
- **Tools ▸ Customize Tools…** (⇧⌘T) arranges tabs, creates your own (for example a "Portraits" tab with just the tools you use for portraits) and saves **workspaces**.
- The **Tools** menu lists every tool by tab; choosing one shows it.
- **Adjustments ▸ Edit All Selected** applies each change to every selected photo.



A custom "Portraits" tab next to the built-in tabs.

The histogram panel

The histogram sits under the tool tabs, above the panels, whichever tab is open. Under it, three checkboxes share one line: **Clipping** (clipped highlights in red and shadows in blue on the image), **Focus peaking** and **Soft proof**. When the tools pane is too narrow for one line, they go on two centred lines. The soft proof options (profile, rendering intent, black point compensation and gamut warning) appear below them only while **Soft proof** is on.

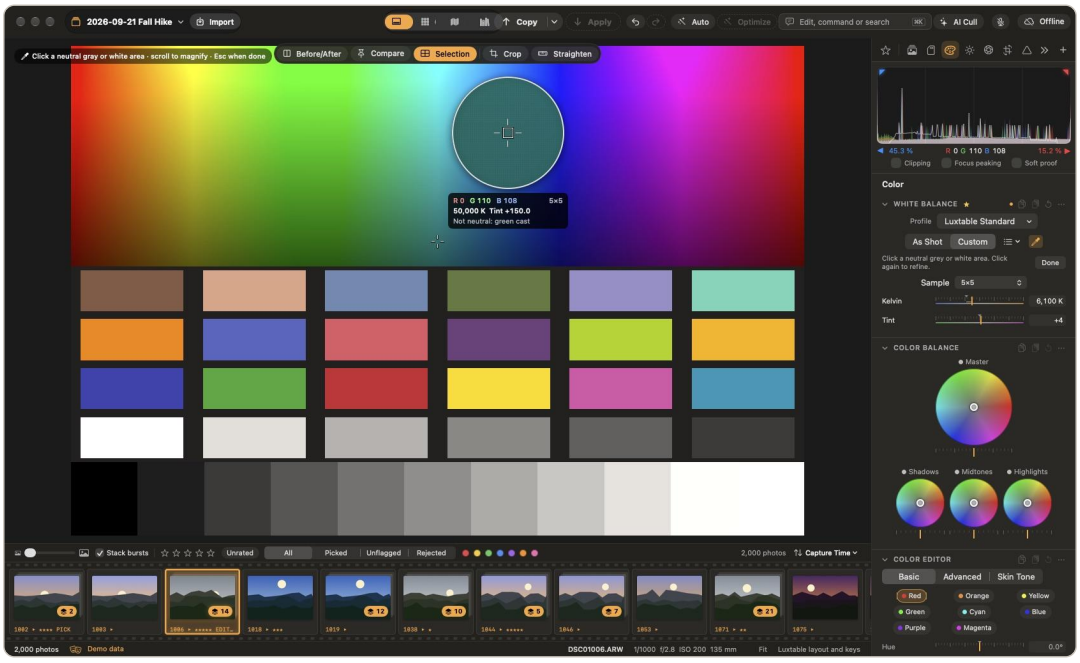
Light

- **Exposure:** exposure, contrast, brightness and saturation.
- **High Dynamic Range:** highlights, shadows, whites and blacks.

- **Levels** and **Curve** (RGB and per channel).
- **Presence:** texture, vibrance and dehaze.

White balance and colour

- **White Balance:** as shot, presets, temperature and tint, or the picker (W): click something that should be neutral.
- The picker's loupe, a magnifying circle beside the cursor, shows exactly what you are about to sample, with a readout below it. It opens zoomed out, showing about 60 pixels across, so you can see where you are picking. Scroll over the loupe to zoom in or out.
- **Color Balance** wheels (master, shadows, midtones, highlights) and the **Color Editor** for individual colour ranges. Saturation is in Exposure and vibrance in Presence.
- **Camera profiles** (looks): Luxtable Standard, Neutral and Vivid, or an imported Adobe DCP profile.
- I is a colour readout picker; click to pin a readout.



The white balance picker's loupe and readout.

Looks, styles and presets

The **Styles & Presets** panel holds one-touch looks: Portrait, Cinematic, film negatives and slides, faded matte, cross-processed, vintage, three black-and-whites, moody, bright and airy, golden hour, dreamy glow, pop, HDR detail, tilt-shift miniature and halation film.

- Each look previews on the photo as you hover, applies in one click as one undo step, and keeps an **Intensity** slider; every setting it made stays editable.
- The **Original** row at the top is the photo as shot, with no adjustments. Hover it to compare, and click it to go back to it in one undo step, so you don't have to undo through every style you tried. ⌘Z brings your edit back.
- After a click, the tiles stay visible, dimmed, while they refresh with the new look.

- Save your own settings as presets, or add looks with plug-ins.

Edit ▸ Match Total Exposures (⌥⇧⌘M) and **Match White Balance to Focused Photo** even out a set of photos.

Detail and lens

- **Sharpening**, **Noise Reduction** (ISO-adaptive, plus AI Denoise; see AI tools) and **Clarity** (clarity and structure).
- **Lens Correction:** lens profiles, distortion, chromatic aberration, and vertical and horizontal keystone. **Vignetting** corrects or adds a vignette.
- **Effects:** an AI **lens blur** (background blur from an on-device depth model; click to set the focus, choose a bokeh shape, or use a tilt-shift band), glow, film grain and halation.

Layers and masks

The **Layers** tab adds local adjustments. Each layer has its own mask and most of the global tools.

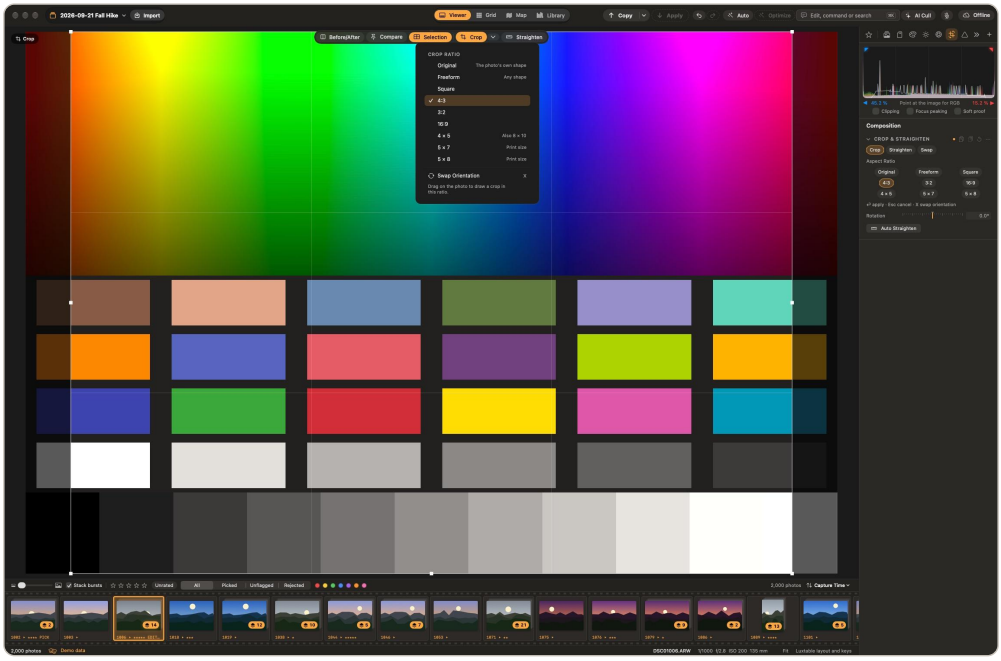
- **Brush** and **erase** strokes, **linear** and **radial gradients**, **luminance** and **colour range** masks. Shapes combine by adding, subtracting or intersecting, with feather and invert. [and] change the brush size. You can bind keys to the mask tools in Settings ▸ Shortcuts.
- **AI masks** for the sky, the subject, and people: a person, or parts of them (face, skin, eyes, teeth, hair, clothes). The sky and people-part models are downloaded the first time you use them.

Heal, clone and remove

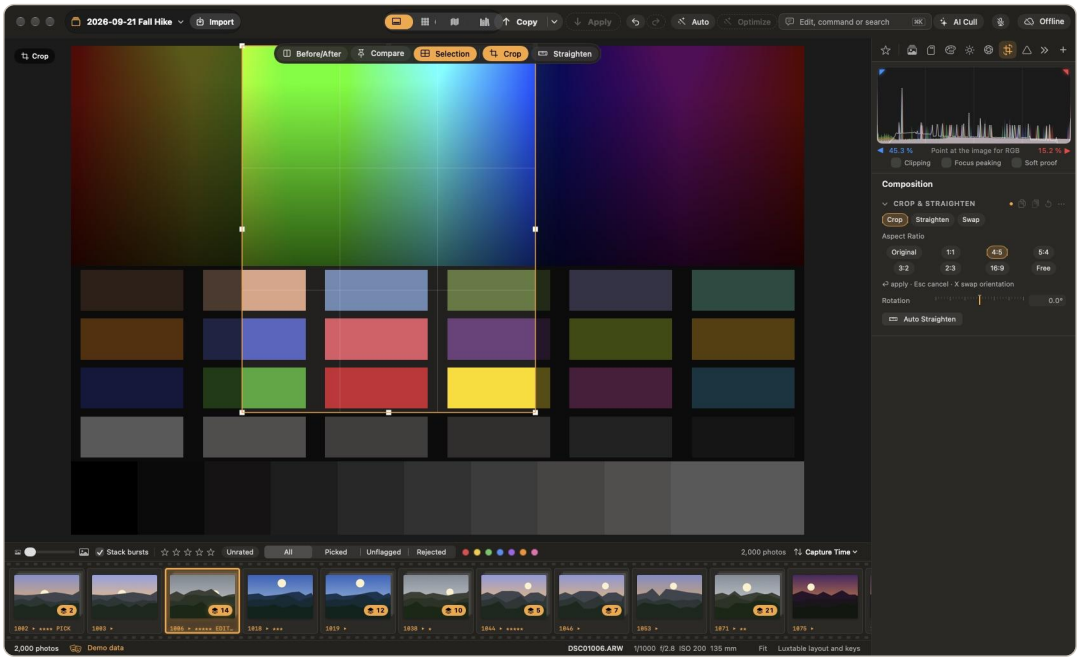
In the **Heal & Clone** panel (Details tab), heal and clone spots such as sensor dust. The **Remove** tool fills whatever you paint over from its surroundings: paint over a distraction, release, and it's gone. A downloaded on-device model (LaMa) does the fill when it is installed; a built-in fill is used until then.

Crop and straighten

- Click **Crop** above the photo (or press C) to open the crop tool. Clicking **Crop** also drops down the crop ratios: **Original** (the photo's own shape), **Freeform**, **Square**, **4:3**, **3:2**, **16:9**, and the print sizes **4 × 5** (also 8 × 10), **5 × 7** and **5 × 8**. The chosen ratio is highlighted. Its ▾ opens the list again while you crop; the Crop panel (Composition tab) has the same ratios.
- Drag on the photo to draw a crop: it keeps the chosen ratio, in the direction you drag (a tall drag gives a portrait crop). Drag the handles to resize it or drag inside it to move it; ↩ applies, Esc cancels, X swaps the orientation.
- The ratio you pick stays chosen, for this photo and every later one, until you pick another, also after Luxtable restarts. Until you choose one it is **Original**. Picking a ratio crops the current photo to the largest box of that ratio inside its current crop; **Freeform** only unlocks the shape.
- R is the straighten tool: drag along something that should be level.
- ⌘[/ ⌘] rotate left or right.



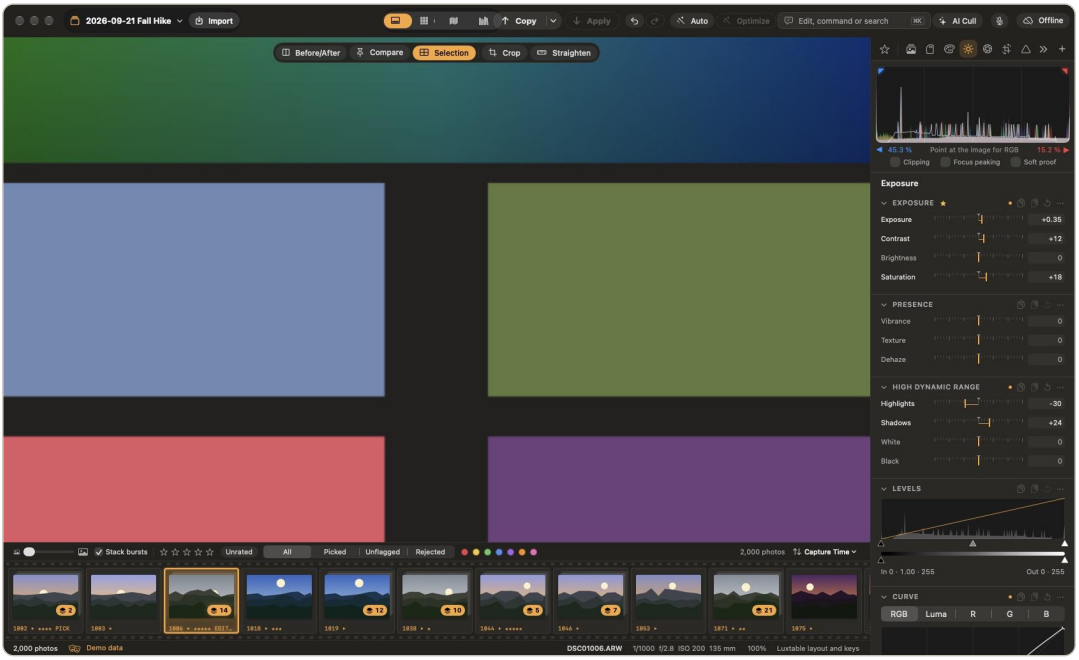
The Crop button's ratio list, with 4:3 chosen.



The crop tool at 4:5, mid-drag.

Variants, history and soft proofing

- **New Variant** (⌘D) makes another version of a photo without duplicating the file.
- **History** (**View ▸ Show History**) lists the photo's edit steps, newest on top. Click a step to go back to it, hover to preview it.
- **Soft Proof** (⇧S, or the **Soft proof** checkbox under the histogram) shows the photo through any installed ICC profile, such as a printer and paper, and can mark out-of-gamut colours.



Full-resolution detail at 100% (Space or ⌥⌘1).

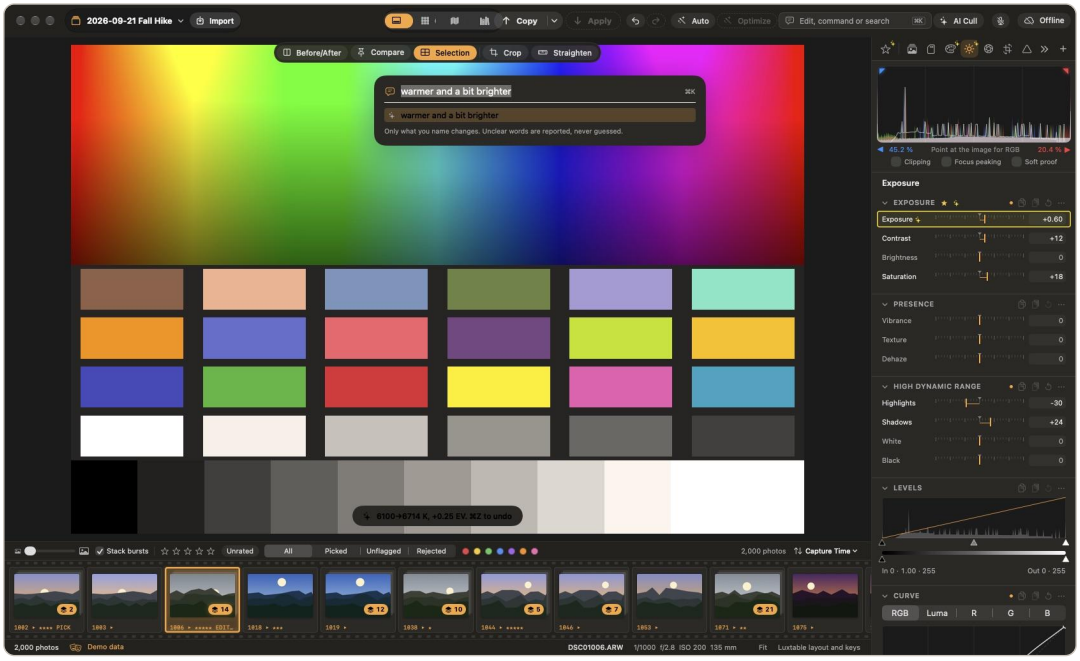
CHAPTER 8

AI tools

Luxtable's AI tools run on your Mac. Some of their models are downloaded once, the first time you use the tool; your photos are never uploaded. Every change an AI tool makes is shown and can be undone on its own.

Edit with words

Press ⌘K and type what you want, for example "warmer and a bit brighter" or "less contrast, more shadows". Luxtable changes only what you name and tells you what it did. Optionally, Settings ▸ AI can send the text (never the photo) to the Jev service with your own API key for phrasing the built-in parser doesn't understand.



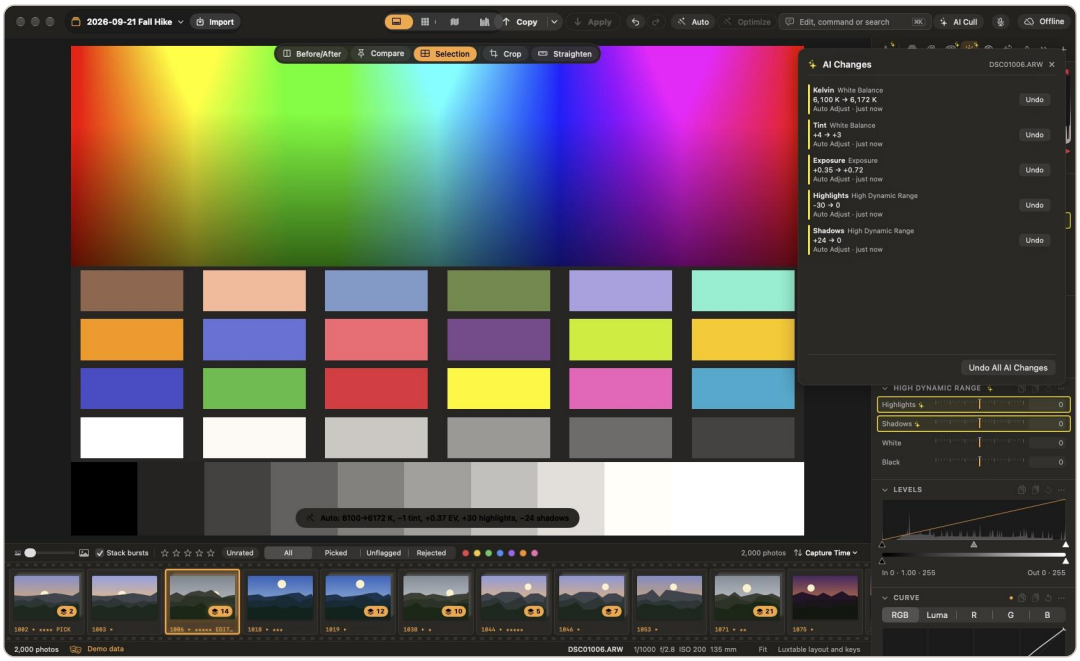
"warmer and a bit brighter", applied.

Auto Adjust and AI Optimize

- **Auto Adjust** (⇧A or ⇧⌘A) sets white balance and tone as a quick starting point.
- **AI Optimize** (⌥⌘A) looks at the photo (the scene, faces, sky, noise) and proposes measured adjustments, each with the reason. Choose which ones to apply.
- **AI Looks** suggests three or four looks that suit the photo (a person, low light, a landscape, a sunset), each with the reason.

AI Changes

Every setting that AI Optimize, Edit with words or Auto Adjust changed gets a yellow outline in the tools. Its tooltip says what it was changed to, from what, and why. **Adjustments ▸ AI Changes…** (⌥⇧⌘A) lists them all with who changed them and when. Undo one, or **Undo All AI Changes**; your own edits made since are kept. The list is saved in the sidecar.



After Auto Adjust: the changed tools are outlined and the AI Changes list shows each change.

AI Denoise and AI Upscale

- **AI Denoise** (Noise Reduction panel) removes noise with a neural network on the Neural Engine and GPU. Turn it on for the selected photos and they are denoised in the background; exports denoise just before they render.
- **AI Upscale** (**Photo ▸ Enhance ▸ Upscale 2×**) makes a new photo twice the size on each side, kept as a linear DNG next to the original with its edit. Export recipes can enlarge with it too. Its model is downloaded the first time, after you agree to its terms.

AI Cull and voice commands are described in Culling, and AI masks and lens blur in Developing.

CHAPTER 9

Copy and apply

Edit one photo, then give the same look to the rest. The **Copy** and **Apply** buttons sit at the top right.

- **Copy** (⇧⌘C) takes the photo's edited adjustments in one click. Its ▾ menu switches to **Custom**, where you tick the groups to copy. Crop, heal and layers are off unless you tick them.
- **Apply** (⇧⌘V) pastes onto every selected photo except the one copied from, as one undo step. The clipboard stays until the next Copy.
- **Copy from Previous** (⌥⌘V) applies the previous photo's adjustments to this one.
- A pasted crop respects each photo's aspect ratio and orientation, and keeps its own rotation and, by default, its straighten.
- Exposure can be pasted as an absolute value or as a relative change.

CHAPTER 10

HDR and panorama

Select two or more photos and choose:

- **Photo ▸ Merge to HDR…** (⇧⌘H): bracketed exposures are aligned, deghosted (anything that moved comes from one frame) and merged in linear light.
- **Photo ▸ Merge to Panorama…** (⇧⌘M): overlapping frames are stitched with a perspective, cylindrical or spherical projection (chosen automatically from the angle covered) and cropped to the largest clean rectangle.

The result is a new linear DNG next to the sources, with the full editing latitude of a RAW. The originals are not changed. Merging needs the RAW helper.

Best Shot

A hand-held burst rarely has one frame where everyone's eyes are open and everything is sharp. **Photo ▸ Merge to Best Shot…** (⇧⌘B) makes that frame. Select the frames of a burst, or one photo of a burst stack for its whole burst, and choose it. You can also right-click a burst in the filmstrip or the grid and choose **Merge to Best Shot…**: that merges that burst's frames, whatever is selected.

The best frame is the base and the others are aligned to it. Each face that is better in another frame, open eyes first, is blended in, and so are regions that are clearly sharper elsewhere. The preview says what was taken: faces improved, sharper regions, and faces that moved too much to swap.

The result is `<name>–Best.dng` next to the sources. It joins the burst and becomes its cover, unless you chose a cover yourself.

CHAPTER 11

Exporting

Select photos and choose **File ▸ Export…** (`⌘E`). The export sheet lists your **recipes**; tick one or several and every selected photo is exported through each, in the background.

Recipes

- **Format:** JPEG, HEIC, 16-bit TIFF, or 8- or 16-bit PNG, with quality.
- **Size** (with or without upscaling, or AI Upscale), output **sharpening** and **colour space** (sRGB, Display P3, or the soft-proof ICC profile with its rendering intent).
- **Destination** and **file name** template.
- **Metadata:** include EXIF, your IPTC fields and keywords, or leave out the GPS position and place names. With metadata included, the star rating and keywords go in too, and **Colour label and pick as keywords** adds tags such as `Luxtable/Label/Red`, `Luxtable/Pick` and `Luxtable/Reject`, which photo servers such as Immich show as tags.
- **Watermark:** a line of text (font, size, colour, opacity, shadow; `{copyright}`, `{year}`, `{filename}` tokens) or a PNG logo, at one of nine positions. It keeps the same relative size at every export size, and the editor previews it on the focused photo.
- **Post-processing:** export steps from plug-ins, for example a border.

Google Photos

A JPEG or HEIC recipe can also upload its files to Google Photos: tick **Also upload to Google Photos** and name an album. Luxtable adds to its own album with that name or creates it, uses each photo's caption as its description, and shows progress per photo and a list of any that failed. Sign in once in Settings ▸ Google Photos; the sign-in is kept in your Keychain. Google Photos is an optional integration: turn it on in Settings ▸ Plugins first. A build without a built-in Google client can use your own (see `docs/GOOGLE_PHOTOS.md`).

CHAPTER 12

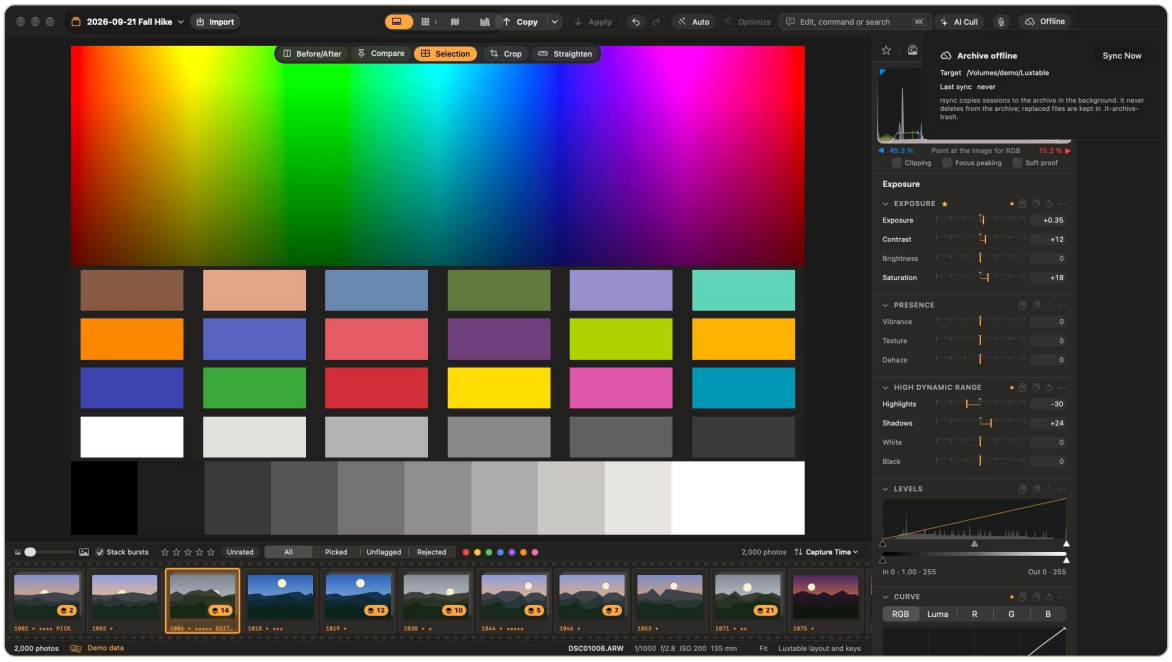
Archive, Immich and Amazon Photos

Archive, Immich and Amazon Photos are optional integrations. Each is off until you turn it on in Settings ▸ Plugins ▸ Integrations, and its Settings pane appears while it is on.

Archive

Luxtable can copy your sessions to an archive in the background: a folder (for example on a NAS) or S3-compatible object storage. Set it up in Settings ▸ Archive or the Import & Archive tab, and tick **Archive sessions in the background**.

- The archive runs when there are changes, after an import and when you close a session, and checks on the interval you choose (**Check every**).
- Choose whether rejected photos are archived too (**Archive rejected photos**). With it off, rejects and the session's Trash stay out of the archive.
- Each session gets its own folder in the archive. Switching destinations never moves or deletes what is already archived.
- The archive button in the toolbar shows the status: up to date, behind, offline or a problem, with the recent runs.



The archive popover after a run.

A folder

Choose **Folder** as the **Destination**, then a folder on a mounted volume (a NAS share or an external drive) or on this Mac. Luxtable copies to it with rsync. Nothing is written while the volume isn't mounted: changes wait, and the archive catches up when it is back. A local folder must already exist (**Create It** makes it), and can't be inside a session.

Object storage

Choose **Object storage** as the **Destination** to archive to any S3-compatible service, such as Amazon S3, Backblaze B2, Wasabi, Cloudflare R2 or MinIO. Fill in:

- **Endpoint:** the service's S3 address. Leave it empty for Amazon S3, where the region picks it.
- **Region**, for example `us-east-1` (Cloudflare R2 uses `auto`).
- **Bucket** and **Path**: the path is a key prefix, so `luxtable/` puts a session at `luxtable/<session>/…`.
- **Path-style addressing**, if the service needs it (MinIO and some others do).
- **Access key ID** and **Secret access key**. They are kept only in your Keychain, never in preferences or files, and the secret is never shown again. **Remove Keys** deletes them.

Save, then **Test Connection**: it checks the bucket and leaves one small test object under the path, in `.lt-archive-test/`.

The endpoint must use `https://`. Plain `http://` is accepted only for a server on your local network (such as a MinIO on your LAN), so your files and keys never cross the internet unencrypted.

Unchanged files are skipped, every upload is verified, and large files go up in parts. Only a consistent snapshot of the session database is archived, never the live one, and caches and half-written files stay out.

Never deletes

The archive never deletes anything. A file that changed is archived again only after its previous version is moved aside, into `.lt-archive-trash/` with the time: in the archive folder, or under the path in the bucket. Developed copies and XMP sidecars for Immich travel with the session.

Immich

Immich is a self-hosted photo server. Luxtable works with an Immich server that indexes your archive folder as an **external library** (mounted read-only is fine): Luxtable never uploads to Immich, and Immich never writes to the archive. Turn Immich on in Settings ▸ Plugins, then set it up in Settings ▸ Immich.

1. **Server:** enter the server address and an API key (create it in Immich under Account Settings ▸ API Keys; it is kept in your Keychain). **Save**, then **Test Connection**.
2. **Where Immich sees the archive:** the archive folder as the server sees it and as this Mac sees it, for example `/mnt/photos` on the server is `/Volumes/photos` on this Mac. **Detect** fills it in from an archived photo Immich already has, or from the library's import path.
3. **Developed copies:** choose the photos you want in Immich as you edited them, then **Photo ▸ Keep a Developed Copy for Immich** (also in the browser's context menu). A JPEG with every edit goes to the session's `Developed/` folder, and the archive carries it to the NAS. It is rendered again when you edit, re-rate or re-tag the photo. A badge on the thumbnail shows the copy (orange when it is out of date). Set

the JPEG quality and size in Settings ▸ Immich. **Photo ▸ Stop Updating Developed Copy** leaves the existing file alone, since nothing deletes from the archive.

4. **XMP sidecars:** with **XMP sidecars next to the originals** on (the default), `IMG_0001.ARW.xmp` next to each archived original carries the stars, title, caption, keywords and position, so RAWs show them in Immich too. With **Colour label and pick as tags** on, the colour label and the pick or reject flag go in as tags as well. A sidecar Luxtable didn't write (from Lightroom or Capture One) is never touched.
5. **Albums:** each album follows a rule, such as a minimum star rating, picked photos only, a colour label or a keyword ("★★★★★" → "Luxtable ★★★★★"), across every archived session. **Add Album** adds one. After each archive run, or **Sync Albums Now**, Luxtable asks Immich to rescan, then adds the matching photos (the developed copy when there is one, otherwise the original) and removes only the photos it added itself. Photos you put in an album by hand stay.

Rate and tag in Luxtable: Immich shows what the sidecars and developed copies say. If you only want the developed JPEGs in Immich, add the exclusion pattern `**/*.ARW` (and your other RAW extensions) to the external library. Album changes and scan requests are the only things Luxtable asks of Immich; scans need an admin's API key (otherwise Immich finds new files on its own schedule).

Immich needs a folder it can index, so it needs the archive to go to a folder. With object storage, album sync is skipped, and Settings ▸ Immich says so.

Amazon Photos hand-off

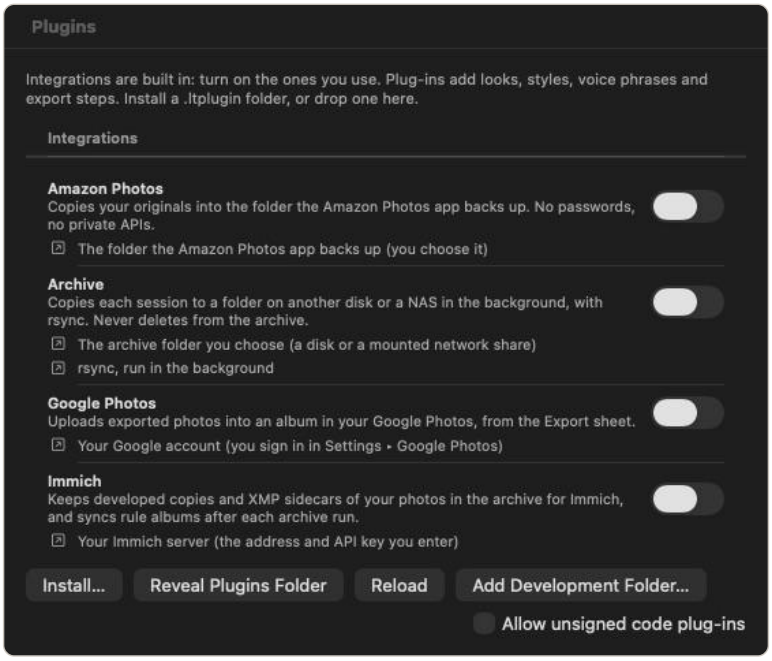
Amazon Photos has no public API, so Luxtable mirrors your originals into a folder that the official Amazon Photos app backs up. Turn it on in Settings ▸ Plugins, then choose that folder in Settings ▸ Amazon Photos, and choose whether rejected photos are sent too. It works with either archive destination.

CHAPTER 13

Plug-ins

Plug-ins add looks, styles, voice phrases, and export steps. Manage them in Settings ▸ Plugins.

- **Install:** drop a `.ltplugin` folder on the pane, or **Install…**. The sheet shows what the plug-in says it does before anything is installed. Turn each plug-in on or off with **Enabled**.
- **Nothing is granted at install.** Looks, styles and voice phrases are data only. An export step is WebAssembly code that runs sandboxed with time and memory limits; code from an unsigned or untrusted author waits for your approval and never gets file, network or account access silently. If a plug-in's files change, it waits for your approval again.
- Luxtable's own looks and styles are a bundled plug-in. Developers can **Add Development Folder…** to work on a plug-in in place; see `docs/PLUGINS.md`.



Settings ▸ Plugins.

Optional integrations

Four integrations are built into Luxtable as bundled plug-ins, so it stays out of services you don't use. Turn each on or off in **Settings ▸ Plugins ▸ Integrations**; its own Settings pane appears while it is on.

INTEGRATION	WHAT IT DOES	WHERE TO SET IT UP
Archive	Copies sessions to a folder or to S3-compatible object storage in the background; never deletes from the archive. Setup offers it.	Settings ▸ Archive
Amazon Photos	Copies originals into the folder the Amazon Photos app backs up.	Settings ▸ Amazon Photos
Immich	Developed copies, XMP sidecars and rule albums for an Immich server that indexes the archive (needs the archive, to a folder).	Settings ▸ Immich

INTEGRATION	WHAT IT DOES	WHERE TO SET IT UP
Google Photos	Uploads a recipe's exports into a Google Photos album.	Settings ▸ Google Photos, then the export sheet

- **Defaults:** each starts off. If you had already set one up before they became optional, it stays on after the update.
- **Turning one off** hides its Settings pane, menus, badges, panels, toolbar button and export-sheet section, and stops its background work (anything already running finishes first).
- Its settings, sign-ins, keys and files are kept, and turning it back on picks up where it left off.
- They can be turned off but not removed.

See Archive, Immich and Amazon Photos and Google Photos for how each works.

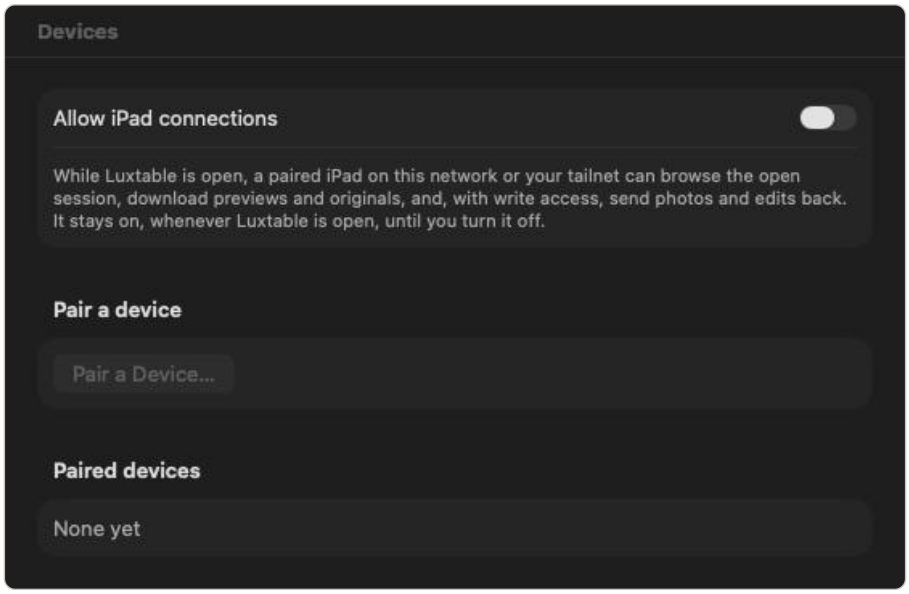
CHAPTER 14

The Mac ↔ iPad link

Settings ▸ Devices lets a paired device work with the session open on your Mac. It is off until you turn on **Allow iPad connections**, and it only runs while Luxtable is open.

- **Pair a Device…** shows a QR code and a 6-digit code. The code works once, for five minutes.
- A paired device on the same network or your tailnet can browse the open session and download previews, originals and sidecars. Turn on **Can send photos and edits** to let it send new photos and edits back; edits merge without losing either side's work.
- Transfers are mutually authenticated (TLS 1.3 with pinned keys), resumable and checksum-verified. **Revoke** removes a device at any time.

> **Note**
>
> The iPad app is planned. Today the `lt-remote` command-line client uses the link; see `docs/REMOTE.md`.



Settings ▸ Devices, with the link off.

CHAPTER 15

Coming from other apps

Luxtable reads other apps' files and never modifies them. Each import starts with a dry run that shows what will change, what is approximate and what is left out, with a checkbox per category, and each can be reverted. Re-importing updates only what is still as imported, never your own changes.

Capture One

- **File ▸ Import Capture One Edits…**: the ratings, colour tags, keywords, picks and supported adjustments of a Capture One session.
- **File ▸ Import Capture One Settings…**: export recipes, the tool workspace, crop ratios, styles and presets, keyword libraries, import naming and the soft-proof profile.
- **Revert Capture One Import…** takes an import back.

Lightroom Classic

- **File ▸ Import Lightroom…**: a Lightroom Classic catalog (`.lrcat`, read from a copy), a folder of XMP sidecars, or Lightroom and Camera Raw presets. Ratings, labels, flags, keywords, titles, captions, collections, virtual copies (as variants), crop, and the develop settings Luxtable has an equivalent for.
- **Revert Lightroom Import…** takes an import back.

Aperture and iPhoto

File ▸ Import Aperture or iPhoto Library… reads an Aperture 3 library or an iPhoto library (iPhoto 6 and later). Choose the library in the sheet; the dry run then lists what will come in.

- **What comes over:** ratings, rejects and hidden photos, flags, colour labels, keywords, titles, captions, places, and projects, events and albums.
- **Edits that map:** exposure, recovery, contrast, definition, saturation, vibrancy, highlights and shadows, white balance on RAW files, straighten, crop and black & white.
- **What doesn't:** levels, curves, retouching, brushed-in adjustments, sharpening, noise reduction, vignettes, faces and smart albums. The dry run lists every one it finds.
- An older iPhoto library kept edits only as rendered copies, so its originals come in unedited.
- The library is only read. Its originals open in place as a session, with Luxtable's data kept outside the library. Originals on a disk that isn't connected are listed until you connect it.
- **File ▸ Revert Aperture or iPhoto Import…** takes the import back. It never touches a Lightroom import, and changes you made in Luxtable afterwards are kept.
- Photos libraries aren't supported. `docs/APPLE_FORMAT.md` has the details of the mapping.

Familiar keys and layout

Settings ▸ Familiar also moves panels, keyboard shortcuts and names to match Lightroom Classic or Capture One.

CHAPTER 16

Settings

Open Settings with ⌘,. Its panes are listed in a sidebar on the left; an integration that is off has no pane until you turn it on in **Plugins**.

PANE	WHAT IT SETS
General	Appearance, the sessions folder, **Open in** (the last session or the Library), opening the import sheet when a card is inserted, your name and copyright for exports, the new-session name, map tiles, and Show Welcome Again
Familiar	Layout, keyboard shortcuts and names like Luxtable, Lightroom Classic or Capture One
Shortcuts	Rebind every single-key command; a key already in use moves to the new command
RAW	Download or remove the RAW helper
Archive	The destination (a folder or S3-compatible object storage), the interval and whether rejects are archived (while the Archive integration is on)
Amazon Photos	The folder the Amazon Photos app backs up (while that integration is on)
Google Photos	Sign in and out; your own Google client if the build has none (while that integration is on)
Immich	The server, API key, path mapping, developed copies, XMP sidecars and album rules (while that integration is on)
AI	How typed edits are interpreted, including the optional Jev service
Voice	Voice commands and your own phrases
Devices	The Mac ↔ iPad link and paired devices
Plugins	The optional integrations (on or off), installed plug-ins, their permissions, and development folders
Updates	Automatic update checks, beta versions and Check Now

CHAPTER 17

Troubleshooting

"Luxtable can't develop this file yet"

Many RAW formats need the RAW helper: click **Get RAW Helper** on the photo, or open Settings ▸ RAW. Fujifilm X-Trans RAFs and Nikon High Efficiency (HE/HE*) NEFs import and cull from their embedded previews, but don't develop yet. A few rare formats (Canon CRW, Sigma X3F, Phase One IIQ, ARRI) are refused. `docs/CAMERAS.md` lists what is tested.

A session or photos show as disconnected or missing

Connect the drive they are on. If you moved the files, use **Locate…** or **Locate Folder…**; files moved by Luxtable itself or renamed are found by content. Nothing is removed while a drive is away.

AI Denoise is unavailable

The app was built with `LT_SKIP_MODELS=1`, or without Python 3.13 or a network connection at build time. Rebuild without that variable.

An AI tool says its model isn't installed

Some models (removal, sky and people masks, depth, photo search) are downloaded the first time you use the tool. Check the network connection and try again; a damaged download is offered again. AI Upscale needs a build that sets a download location.

The map is blank

Map tiles may be turned off: choose **Show Map**, or turn them on in Settings ▸ General. Photos without a location don't appear until you drag them onto the map.

The archive says it is offline or behind

The archive drive or NAS isn't mounted, or the object storage service didn't answer. Syncs wait until it is back, and nothing in the archive is ever deleted. For object storage, **Test Connection** in Settings ▸ Archive says what is wrong; an `http://` endpoint outside your local network is refused (use `https://`).

The Archive, Immich, Amazon Photos or Google Photos settings are missing

That integration is off. Turn it on in Settings ▸ Plugins ▸ Integrations; its settings were kept.

Immich doesn't show my albums or ratings

Check that the archive goes to a folder (Immich can't index object storage), that the path mapping in Settings ▸ Immich is right (**Detect** fills it in), and that Immich's external library points at the archive. Without an admin API key, Immich finds new files only on its own schedule.

macOS keeps asking for permissions after each build

Run `scripts/make-signing-identity.sh` once and rebuild, so every build is signed with the same identity.

The keys don't do what this guide says

Settings ▸ Familiar may be set to Lightroom Classic or Capture One keys, or a key may have been rebound in Settings ▸ Shortcuts. Press `?` to see the keys in use.

Reporting a problem

Open an issue at github.com/sbaillar/luxtable. Report security issues privately, as described in `SECURITY.md`.

APPENDIX

Keyboard shortcuts

These are Luxtable's default keys. Single keys can be changed in Settings ▸ Shortcuts, and Settings ▸ Familiar switches to Lightroom Classic or Capture One keys. Press ? in the app to see the keys in use.

Navigate and view

KEYS	DOES
← → ↑ ↓	Previous / next photo, row up / down
⇧ + arrows	Extend the selection
⌘A	Select all photos
G / E	Grid / viewer
↩	Open in the viewer
F or Tab	Hide panels
B	Hide the filmstrip
^⌘= / ^⌘-	Larger / smaller filmstrip thumbnails
^⌘F	Full screen
Space	100% at the cursor
⌘+ / ⌘-	Zoom in / out
⌥⌘0 / ⌥⌘1	Zoom to fit / actual size
Y	Before / after
K	Pin for compare
S	Expand / collapse stack (turns burst stacking on if it is off)
← → / ↩ / Esc	In the burst strip: previous / next frame, use as the stack's cover, close
N or ⌥⌘S	Survey the selection
J or ⌥⌘J	Clipping warning
⌥⌘P	Focus peaking
⌥⌘L / ⌥⌘N	Loupe / navigator
⇧S	Soft proof on / off
⌘0	Show all photos
⌘1 – ⌘5	Show only that many stars
⌥⌘[/ ⌥⌘]	Move tool tab left / right
^⌘S	Simple mode / full tools
Esc	Back / close
? or ⌘/	Keyboard cheat sheet

Cull

KEYS	DOES
1 – 5	Rate 1–5 stars
0	Clear rating
6 7 8 9 -	Red, yellow, green, blue, purple label
P / X / U	Pick / reject / unflag
⇧⌘K	AI Cull
⌘D	New variant
^⌘V	Voice commands on / off

Edit

KEYS	DOES
C	Crop tool
R	Straighten tool
W	White balance picker (scroll over its loupe to zoom)
I	Colour readout picker
[/]	Brush size
⌘[/ ⌘]	Rotate left / right

KEYS	DOES
⌘Z / ⇧⌘Z	Undo / redo
⇧⌘R	Reset all adjustments
⇧⌘C	Copy adjustments
⇧⌘V	Apply to the selection
⌥⌘V	Copy from previous
⌥⇧⌘M	Match total exposures
⇧A or ⇧⌘A	Auto Adjust
⌥⌘A	AI Optimize
⌥⇧⌘A	AI Changes
⌘K	Edit with words
⇧⌘H / ⇧⌘M	Merge to HDR / panorama
⇧⌘B	Merge to Best Shot
⇧⌘T	Customize tools

Sessions and files

KEYS	DOES
⇧⌘N	New session
⌘O	Open a session or folder
⌥⌘W	Close session
⇧⌘L	Browse all sessions in the Library
⇧⌘I	Import from card
⌘E	Export
⌘F	Find
⌘,	Settings